1. Introduction to HTML4 & Document Structure

> **Definition: HTML (HyperText Markup Language) version 4 is the standard markup language used to create and structure the content of web pages. It uses a system of tags to tell the web browser how to display text, images, and links.**

HTML is not a programming language; it is a markup language. This means it doesn't have logic, loops, or variables. Instead, it uses "tags" enclosed in angle brackets (like `<html>`) to annotate text, images, and other content for display in a web browser.
HTML4 was released in 1997 and became the core standard for web development before HTML5 was introduced. It strictly separates the structure of a document from its presentation (which is handled by CSS). An HTML document is saved with a **.html** or **.htm** extension.

Basic Structure of an HTML4 Document:

Every valid HTML page must follow a specific skeleton consisting of four main tags:

- **<html> ... </html>:** The root element that wraps all the content on the entire page.
- **<head> ... </head>:** Contains meta-information about the document, such as its title, character set, and links to CSS files. This content is *not* displayed directly on the webpage.
- **<title> ... </title>:** Specifies the title of the webpage, which appears in the browser's title bar or tab. It must be placed inside the `<head>` section.
- **<body> ... </body>:** Contains all the visible content of the webpage, such as headings, paragraphs, images, tables, and links.

Key Points:

- HTML tags usually come in pairs: an opening tag `<tag>` and a closing tag `</tag>`.
- Some tags are "empty" or "void" elements, meaning they don't have a closing tag (e.g., `<br>` for line break, `<hr>` for horizontal rule).
- HTML tags are not case-sensitive (e.g., `<BODY>` is the same as `<body>`), but lowercase is highly recommended and standard practice.

```
<!-- Basic HTML4 Skeleton --> <html> <head> <title>My First Webpage</title> </head> <body> <h1>Welcome to
my website!</h1> <p>This is a paragraph of text.</p> </body> </html>
```

> 👉 **Hinglish Explanation:** HTML web pages ka skeleton (haddiyon ka dhancha) banata hai. Ye koi programming language (jaise C/C++) nahi hai. Isme hum 'Tags' ka use karte hain. Sabse pehle `<html>` lagate hain, phir `<head>` jisme page ka title hota hai jo tab par dikhta hai. Asli content (jo screen par dikhta hai) hamesha `<body>` tag ke andar likha jata hai.

2. Important HTML4 Tags & Elements

Definition: An HTML element is defined by a start tag, some content, and an end tag. Attributes provide additional information about HTML elements and are always specified in the start tag.

To create a functional webpage, you must know how to format text, create lists, add hyperlinks, and insert images. Here are the core HTML4 tags categorized by their purpose:

1. Text Formatting Tags:

- **Headings:** `<h1>` to `<h6>`. `<h1>` is the largest and most important heading, while `<h6>` is the smallest.
- **Paragraphs:** `<p>` defines a paragraph of text. The browser automatically adds empty space before and after it.
- **Formatting:** `<b>` for **bold** text, `<i>` for *italic* text, `<u>` for <u>underlined</u> text.

2. Lists (Ordered & Unordered):

Lists are used to group related items.

- **Unordered List (Bullet points):** Created using the `<ul>` tag. Each list item is defined with the `<li>` tag.
- **Ordered List (Numbered):** Created using the `<ol>` tag. Each item is defined with the `<li>` tag.

3. Links and Images:

- **Hyperlinks:** The Anchor tag `<a>` creates a clickable link to another page. The `href` attribute specifies the destination URL.
- **Images:** The `<img>` tag embeds an image. It requires two main attributes: `src` (the path/URL of the image file) and `alt` (alternative text shown if the image fails to load). It is an empty tag (no closing tag).

4. Division / Layout Tag:

The `<div>` tag is a block-level element used as a container to group HTML elements together. In HTML4, it is heavily used to divide the webpage into sections for CSS styling.

```
<!-- Example of Lists, Links, and Images --> <h2>My Favorite Subjects</h2> <ul> <li>IT Systems</li>
<li>Applied Physics</li> </ul> <!-- Link with href attribute --> <a href="https://rgpv.ac.in">Visit RGPV
Website</a> <!-- Image with src and alt attributes --> <img src="college_logo.jpg" alt="College Logo">
```

👉 **Hinglish Explanation:** Text ko mota karne ke liye `<b>`, headings ke liye `<h1>` se `<h6>` use karte hain. `<a>` tag link banane ke kaam aata hai jisme 'href' me hum website ka address daalte hain. Image lagane ke liye `<img>` tag use hota hai jiska closing tag nahi hota, usme 'src' (source) dena zaroori hai. `<div>` ek dabbe (box) ki tarah hai jiske andar hum aur tags ko group karte hain.

3. HTML Tables and Forms

Definition: Tables are used to arrange data into rows and columns. Forms are interactive sections of a webpage used to collect user input (like login details or feedback) and send it to a server.

HTML Tables:

Tables are created using the `<table>` tag. Inside the table, rows are created using `<tr>` (Table Row). Inside rows, heading cells are created using `<th>` (Table Heading, default bold and centered), and standard data cells are created using `<td>` (Table Data).

HTML Forms:

The `<form>` tag is a container for different types of input elements.

- **<form action="url" method="GET/POST">:** The `action` attribute specifies where to send the form data upon submission. The `method` specifies how to send it (POST is used for sensitive data like passwords).
- **<input>:** The most used form element. Its `type` attribute dictates its behavior.
  - `type="text"`: A single-line text input field.
  - `type="password"`: Hides characters (shows asterisks/dots).
  - `type="radio"`: Radio button (for selecting only ONE option from many).
  - `type="checkbox"`: Checkbox (for selecting MULTIPLE options).
  - `type="submit"`: A button to submit the form data.
- **<textarea>:** For multi-line text input (like an address or comment).

```
<!-- TABLE EXAMPLE --> <table border="1"> <tr> <th>Name</th> <th>Marks</th> </tr> <tr> <td>Rahul</td>
<td>85</td> </tr> </table> <!-- FORM EXAMPLE --> <form action="submit.php" method="POST"> Username: <input
type="text" name="user"> <br> Password: <input type="password" name="pass"> <br> <input type="submit"
value="Login"> </form>
```

👉 **Hinglish Explanation:** Table banane ke liye `<table>` use hota hai, `<tr>` se line (row) banti hai aur us line ke andar data dalne ke liye `<td>` use karte hain. Form ka use data lene ke liye hota hai (jaise Facebook login page). `<input type="text">` se dabba banta hai jisme hum likh sakte hain. Password field me type="password" dene se letters dot (****) ban jate hain taaki koi dekh na sake.

4. Introduction to CSS (Cascading Style Sheets)

Definition: CSS is a style sheet language used for describing the presentation, design, and layout of a document written in HTML. It controls colors, fonts, spacing, and positioning.

If HTML is the skeleton of a webpage, CSS is the skin, clothes, and makeup. Without CSS, web pages would look like plain, boring black-and-white text documents from the 1990s.

Advantages of CSS:

- **Separation of Concerns:** Keeps HTML structure clean by separating design code from content code.
- **Saves Time:** You can write CSS once and apply it to multiple HTML pages. Changing one CSS file updates the design of the entire website.
- **Faster Page Load:** Less code duplication means smaller file sizes, resulting in faster loading speeds.
- **Responsive Design:** CSS can change the display rules depending on screen size (mobile vs. desktop).

CSS Syntax:

A CSS rule-set consists of a **Selector** and a **Declaration Block**.

- **Selector:** Points to the HTML element you want to style (e.g., `h1`, `p`, or a specific class).
- **Declaration Block:** Enclosed in curly braces `{}`. It contains one or more declarations separated by semicolons. Each declaration includes a CSS **Property** name and a **Value**, separated by a colon.

Three Ways to Insert CSS:

1. **Inline CSS:** Added directly inside the HTML tag using the `style` attribute. *(Not recommended for large pages)*.
   `<p style="color: red;">Text</p>`
2. **Internal CSS:** Placed inside a `<style>` tag within the `<head>` section of the HTML document. *(Good for single-page styling)*.
3. **External CSS:** Written in a separate file with a `.css` extension and linked to the HTML page using the `<link>` tag in the head section. *(Best practice)*.

```
CSS SYNTAX DIAGRAM:

 Selector      Property    Value
   |              |          |
   v              v          v
  h1 {          color :     blue ;   }
                 \_______ ______/
                         v
                  Declaration Block
```

👉 **Hinglish Explanation:** HTML sirf structure deta hai, par usko sundar (beautiful) banane ka kaam CSS ka hai. CSS me hum 'Selector' se target karte hain ki kis tag par design lagana hai (jaise `h1`). Phir bracket `{}` ke andar Property (jaise color) aur uski Value (jaise red) dete hain. Sabse accha tareeka External CSS hai, jisme ek alag .css file banakar saari websites par ek sath design lagaya ja sakta hai.

5. Core CSS Properties & The Box Model

Definition: CSS properties dictate how elements appear. The CSS Box Model is a fundamental concept representing how every HTML element is wrapped in a rectangular box consisting of margins, borders, padding, and the actual content.

The CSS Box Model:

Every HTML element can be considered as a box. Understanding the box model is crucial for layout and spacing. From inside out, the box model consists of:

1. **Content:** The actual text or image area.
2. **Padding:** Clears an area *inside* the border, around the content. It is the inner space.
3. **Border:** A visible line that wraps around the padding and content.
4. **Margin:** Clears an area *outside* the border. It separates this box from other elements. It is the outer space.

```
+----------------------------------------+
|                MARGIN                  |
|   +----------------------------------+ |
|   |            BORDER                | |
|   |   +-------------------------+    | |
|   |   |         PADDING         |    | |
|   |   |   +-----------------+   |    | |
|   |   |   |    CONTENT      |   |    | |
|   |   |   |   (Text/Image)  |   |    | |
|   |   |   +-----------------+   |    | |
|   |   +-------------------------+    | |
|   +----------------------------------+ |
+----------------------------------------+
```

Common CSS Properties:

- **Colors & Backgrounds:** `color` sets text color. `background-color` sets the background. Values can be names (red), HEX codes (#ff0000), or RGB.
- **Fonts:** `font-family` (e.g., Arial, sans-serif), `font-size` (e.g., 16px or 1.2em), `font-weight` (e.g., bold).
- **Text Alignment:** `text-align: center;` centers the text inside its container.
- **Borders:** Defines thickness, style, and color. `border: 2px solid black;`

> 👉 **Hinglish Explanation:** CSS Box model web design ka sabse important topic hai. Har tag ek dabba (box) hota hai. Content aur border ke beech ki khali jagah ko **Padding** kehte hain (andar ki space). Aur border ke bahar ki khali jagah ko **Margin** kehte hain (do dabbon ko door rakhne ke liye). CSS me rang (color), text size aur background change karna bahut aasan hai, bas proper property name pata hona chahiye.

6. Making a Basic Personal Webpage

> **Definition: A personal webpage is a basic website created to showcase an individual's profile, skills, and contact information. It integrates HTML4 for structural layout and CSS for visual styling.**

In web development exams, you are often asked to write the code for a basic personal webpage. The approach involves two steps: first creating the skeleton using `<div>`, `<h1>`, and `<p>` tags, and then styling it using Internal CSS inside the `<head>` section.

Features included in this example:

- A clean Header section with the person's name.
- An 'About Me' section using paragraph tags.
- A 'Skills' section using an Unordered List.
- A 'Contact' section using a hyperlink.
- CSS styling for backgrounds, fonts, centering, and a card-like layout using margins and padding.

```
<!-- Complete Code for a Basic Personal Webpage --> <html> <head> <title>My Personal Webpage</title>
<style> /* CSS Styling Starts Here */ body { font-family: Arial, sans-serif; background-color: #f4f4f4;
```

```
margin: 0; padding: 20px; } .profile-card { background-color: white; width: 500px; margin: 0 auto; /*
Centers the box */ padding: 30px; border: 2px solid #ccc; border-radius: 10px; text-align: center; } h1 {
color: #2563eb; } ul { text-align: left; } </style> </head> <body> <div class="profile-card"> <h1>Hello, I
am Satyam</h1> <h3>Engineering Student</h3> <hr> <p>Welcome to my personal webpage. I am passionate about
coding, web development, and learning new technologies.</p> <h3>My Skills:</h3> <ul> <li>HTML4 & CSS</li>
<li>C Programming</li> <li>Database Management</li> </ul> <br> <a href="mailto:satyam@email.com">Click here
to Email Me</a> </div> </body> </html>
```

👉 **Hinglish Explanation:** Exam me agar "Create a personal webpage" aaye, toh upar diya gaya code likhna hai. Isme humne Internal CSS use karke ek sundar box (.profile-card class) banaya hai. `margin: 0 auto;` property box ko screen ke center me set kar deti hai. HTML ke simple tags (h1, p, ul, li, a) use karke content likha gaya hai. Ye code minimum 7-8 marks easily dilwa sakta hai!

🔁 Quick Revision Notes (Exam Saver)

- **HTML Basic Structure:** Must include `<html>`, `<head>` (for title/css), and `<body>` (for visible content).
- **Key Tags:** `<h1>` to `<h6>` for headings, `<p>` for paragraphs, `<a href="...">` for links, `<img src="...">` for images.
- **Lists:** `<ol>` creates numbered lists. `<ul>` creates bulleted lists. Both use `<li>` for items.
- **Tables:** `<table>`, `<tr>` (Row), `<th>` (Heading), `<td>` (Data).
- **CSS Meaning:** Cascading Style Sheets. Separates design from structure. Advantages: Saves time, fast loading, responsive.
- **CSS Box Model:** Margin (outer space) -> Border -> Padding (inner space) -> Content.
- **CSS Types:** Inline (in tag), Internal (in head), External (separate .css file).

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